Shared Dictionary Co-Optimized Model-Driven Deep Network for Cross Scene Hyperspectral Image Change Detection

Tongzhen Zhang, *Student Member, IEEE*, Jiahui Qu, *Member, IEEE*, Yunsong Li, *Member, IEEE*, Qian Du, *Fellow, IEEE*, and Wenqian Dong, *Member, IEEE*

***Abstract*—Hyperspectral image (HSI) change detection (CD) aims to accurately detect land cover changes on the Earth's surface, which is crucial in remote sensing applications. Due to the high cost of labor and difficulty of field surveys, a large amount of labeled data for newly captured scenes are hard to attain. Although cross scene CD methods are developed to mitigate the issue of data insufficiency, they are based on "black-box" design that directly learns the input–output mapping. To address this, we propose a shared dictionary co-optimized model-driven deep network (SDCO-MDDN) for cross scene HSI CD. This method learns the representations of HSI from source and target domains in an interpretable manner to reduce domain discrepancy, thus achieving CD with limited labeled samples. The shared dictionary co-optimized (SDCO) model is proposed to obtain the representations of HSI in different domains mapped in a shared subspace. The SDCO model is unfolded into an iterative algorithm, and the SDCO-MDDN is built using this algorithm with carefully designed modules. A global self-correlation matrix is formed to capture the global correlation information of the entire input data, helping SDCO-MDDN focus on highly correlated information. The representations of multitemporal HSIs learned by SDCO-MDDN in an interpretable manner are subtracted in each domain, respectively, and the difference features in each domain are aligned to transfer the knowledge of source domain for CD of target domain. Experimental results show that the proposed method outperforms the state-of-the-art CD methods. The code is available at https://github.com/Jiahuiqu/SDCO-MDDN**

***Index Terms*—Change detection (CD), cross scene, hyperspectral image (HSI), interpretable.**

I. Introduction

CHANGE detection (CD) is a critical remote sensing task that analyzes the images taken at different times in the same area to identify the changes of land-cover objects. Since accurate land surface change acquirement is of great significance to monitor the dynamical development of the Earth's surface ecosystem and human activities, CD has been widely used in many applications, such as urban development [1], disaster assessment [2], and resource utilization and monitoring [3]. In recent years, satellites with hyperspectral sensors have been launched consecutively to acquire hyperspectral data for Earth observation. Hyperspectral image (HSI) with hundreds of narrow spectral bands contains richer spectral information than traditional multispectral images, thus providing potential to achieve finer CD [4], [5]. Therefore, a great number of researches have been developed for HSI CD. Although impressive progress has been made on HSI CD, there are still critical challenges in improving the performance of CD methods. For the most popular deep learning-based methods, sufficient labeled samples are required to obtain high detection accuracy. However, it is time-consuming and human labor sensitive to acquire labeled samples of HSI, which hinders the further development in practical CD applications [6]. Meanwhile, different HSIs may record similar scenes with the same classes of land cover, and the same labeled class from similar scenes shares similar spectral signature. An intuitive idea for improving the performance of CD with limited labeled samples is to utilize the information of the image with sufficient labeled samples from the similar scene.

The straightforward choice is directly using the labeled samples of one scene (source domain) to train the CD model for a new scene with limited labeled samples (target domain). However, the spectral signatures of the same objects undergo alterations due to various factors, including atmospheric, illumination, and weather conditions, which results in domain shift between two domains [7]. Ignorance of domain shift leads to poor CD performance on target domain.

In order to address the aforementioned problem, the domain adaption methods have been developed in recent years [8], [9]. Domain adaption aims to align the distribution between two domains, thus obtaining the adaptative model for target domain CD by exploiting shared knowledge from source domain. In [10], maximum mean discrepancy (MMD) is adopted to minimize the domain shift, thus learning the transferrable features for image classification. Aiming at transferring task-specific features from source to target domains, Huang et al. [11] introduced multikernel MMD (MK-MMD) to learn

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Tongzhen Zhang, Jiahui Qu, Yunsong Li, and Wenqian Dong are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: tzzhang_1@stu.xidian.edu.cn; jhqu@xidian.edu.cn; ysli@mail.xidian.edu.cn; wqdong@xidian.edu.cn).

Qian Du is with the Department of Electronic and Computer Engineering, Mississippi State University, Starkville, MS 39759 USA (e-mail: du@ece.msstate.edu).

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stronger feature representation for SAR target recognition. The joint correlation alignment (CORAL)-based graph neural network [12] proposed domainwise and classwise coral to achieve joint distribution adaptation performance for HSI classification. The stagewise domain adaptation framework [13] was designed to address the domain shift for remote sensing image road segmentation via generative adversarial networks. Li et al. [14] proposed a semisupervised segmentation method of remote sensing image, which adopts GAN to fit the distribution of both labeled and unlabeled data. Nevertheless, few efforts have been made for remote sensing image CD, especially for HSI. In [15], a transferred deep learning-based CD framework was designed for remote sensing image CD, in which lower layers of the network are shared to minimize the feature-level distribution discrepancy. A deep Siamese domain adaptation convolutional neural network (DSDANet) [16] was proposed to match the distribution of the two domains, so as to learn transferrable feature representation. Aiming at taking the prior knowledge of multiple source domains, Yang et al. [17] introduced a CD framework based on selective adversarial adaptation to achieve the multisource and target domains adaptation by two domains discriminators. Qu et al. [18] proposed a feature mutual representation-based graph domain adaptive network (FGDANet), which constructs a pseudo Siamese backbone to improve the feature expression capability. However, the existing domain adaptive CD methods are typically based on black-box neural networks learning an end-to-end mapping from the input to the output. Although these black-box based methods have demonstrated promising performance, they are lack of interpretability, which hinders the understanding of the underlying model mechanisms and the analysis of the roles of different models.

To address the aforementioned issues, we proposed shared dictionary co-optimized model-driven deep network (SDCO-MDDN) for cross scene HSI CD. The proposed method builds an interpretable network to jointly learn the representations of HSI in the source and target domains, which can minimize the domain shift in different domains to achieve CD of target domain. Specifically, the shared dictionary co-optimized (SDCO) model is constructed to jointly learn the dictionaries of source and target domains, which ensures the obtained representation of HSI in different domains is mapped in a shared subspace. The proximal gradient descent (PGD) algorithm is adopted to transform the model solving process into an iterative process of two subproblems. The SDCO-MDDN is constructed by performing each operation in the algorithm with carefully designed modules, thus learning the representation of HSI in an interpretable manner. The difference features are obtained by subtracting the learned representations of multitemporal HSIs, and the distributions of difference features of two domains are aligned to transfer the knowledge of source domain for CD of target domain. The main contributions of this article can be summarized as follows.

1) We propose an SDCO model for cross scene HSI CD, which jointly learns the representations of HSI of source and target domains in an interpretable manner to alleviate the domain discrepancy, so as to achieve CD of target domain with limited labeled samples.
2) An SDCO-MDDN is constructed to exact corresponding relationship between its modules of both domains and the operation of the proposed SDCO-MDDN, thus obtaining the representations in source and target domains in a shared subspace.
3) A global self-correlation matrix is proposed to model the global correlation of the entire input representations, which guides the subnetwork in SDCO-MDDN to better integrate the global information of multitemporal HSIs.

The remainder of this article is structured as follows. The related works is briefly introduced in Section II. In Section III, we introduce the proposed SDCO-MDDN for cross scene HSI CD. Section IV presents the experiments and results analysis. Section V concludes our work in this article.

II. Related Work

A. Traditional CD

Traditional CD methods are divided into four categories, i.e., algebra-based methods, transformation-based methods, classification-based methods, and advanced methods.

Algebra-based methods perform algebraic operations on multitemporal data to generate a change map, including image differencing, image regression [19], image ratio, and change vector analysis (CVA). CVA [20] is a widely used CD method, which distinguishes changed/unchanged regions according to the strength and direction characteristics of multitemporal images. However, the algebra-based approaches are sensitive to disturbances and particularly in the presence of brightness and contrast variations across multiple temporal images.

Transformation-based methods have been proposed to solve this problem, which improve the detection performance by transforming the images to a specific feature space. Popular transformation-based methods include principal component analysis (PCA) [21], multivariate alteration detection (MAD) [22], and slow feature analysis (SFA) [23]. PCA effectively reduces redundant information of images and aggregates the majority of change information. MAD employs a covariance matrix to characterize relationships among multiple variables. However, transformation-based methods are only capable of identifying changed regions.

Classification-based CD can obtain changed regions as well as the "from-to" information. Bruzzone and Serpico [24] proposed a supervised nonparametric approach to explicitly identify land-cover transitions in multitemporal images. Achour et al. [25] proposed an automatic CD method that integrates percent difference (PD) and PCA. Classification-based CD methods have unique advantages because they are fundamentally simple and easy to understand, but the accuracy of CD decreases as it accumulates the classification error in each image.

Traditional methods are easy to understand as they usually rely on explicit algorithms and models, and they may perform well for certain scenarios and problems. However, the manually extracted features of traditional methods are difficult to capture intricate information of multitemporal images.

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Therefore, the traditional methods lack universality across scenarios and are unstable with complex scenes or data, resulting in degraded performance.

B. Deep Learning-Based CD

Over the past few years, deep learning methods have been utilized for HSI CD [26]. Zhang et al. [27] proposed a dual-branch Siamese spatial–spectral transformer attention network, which combines the advantages of CNN and transformer and designs a random weight factor to balance the contributions of the two branches. In order to mine the inherent similarity within hyperspectra, Shi et al. [28] learned a robust difference signature from multiple datasets and leveraged their synergy. Cheng et al. [29] introduce the first spatial–spectral interaction paradigm, incorporating a background-guided prototype autoencoder to enhance the spatial and spectral feature representation of HSIs. A cascade cross-scale network (CCSNet) with a Siamese structure was developed in [30] to extract multiscale bitemporal features for the state-of-the-art performance. Zhang et al. [31] proposed a spectrum-aware transformer network (SATNet) to extract the spectral dependence of HSI through a transformer to obtain more discriminative features. Wang et al. [32] proposed a triplet transformer framework based on parents-temporal attention and brother-spatial attention for HSI CD to fully mine the spatial–spectral and temporal information. Huang et al. [33] introduced a hypermatch training schedule to enhance feature extraction by leveraging both labeled and unlabeled samples. Li et al. [34] proposed an end-to-end cross-band 2-D attention network (CBANet) for HSI CD, which embeds a 2-D spatial–spectral attention module into a cross-ban feature extraction to extract spectral differences. Yang et al. [35] enhance hyperspectral CD by amplifying critical variations through Laplacian sharpening convolution and constructing discriminative difference graphs. Zhang et al. [36] proposed the first intermodal and cross-modal self-supervised dual-branch learning framework (ICSF), which extracts discriminative features through cross-modal contrastive learning effectively addressing the challenge of domain shift. Dai et al. [37] pioneered a domain adaptation and disentangled representation network that explicitly separated domain-invariant content features from domain-specific style features, thereby reducing domain gaps in heterogeneous CD.

Deep learning models have strong capability in dealing with different types of data, making it adaptable to a wide range of applications. However, the aforementioned methods generally exhibit the limitations of noninterpretable. These "black-box" methods include unknown internal decision processes of the models, thus reducing confidence of the results. To address this issue, we take the advantages of both traditional and deep learning methods to enhance the interpretability of the domain adaptation model for CD, rendering its clear physical interpretations for understanding.

III. Proposed Method

The overall framework of the proposed method is shown in Fig. 1, and the details of the proposed method will be introduced in this section. First, we give an overview of the model-driven cross scene HSI CD method in Section III-A, which proposes the SDCO model to map the representations in the same subspace jointly for knowledge transfer and builds the SDCO-MMDN network according to the SDCO model to learn the aligned difference features for CD. Then, the SDCO-MMDN network is described in detail in Section III-B, in which all subnetworks of both domains correspond to the operations in the optimization algorithm to have clear physical meanings, thus obtaining the representations of source and target domains in an interpretable manner. Finally, we introduce the loss of the SDCO-MDDN in Section III-C.

A. Dictionary Co-Optimized Cross Scene HSI CD

Due to the nonnegligible domain shift between source and target domains, directly using the features extracted in source domain for target domain CD will result in severe performance degradation. Although some cross scene methods are developed to alleviate the domain discrepancy, these methods are based on a "black-box" design with limited interpretability. To address this problem, we propose a dictionary co-optimized model for cross scene HSI CD, which fits the aligned representations of two different domains in an interpretable manner for target domain CD.

Specifically, we propose an SDCO model for cross-scene HSI CD. Assuming the existence of discriminative features shared across domains, dictionary learning is employed to jointly learn source and target dictionaries, mapping both into a unified low-dimensional subspace and alleviating domain shift. The coefficient vectors in this subspace serve as learned feature representations. As shown in Fig. 1, multitemporal HSIs are first segmented into superpixels by SLIC to integrate spatial–spectral information with high similarity. These superpixels are then fed into the SDCO model in batches, enabling global information awareness and more accurate dictionary learning. The SDCO model further decomposes segmented HSIs of each domain into a common dictionary and two low-dimensional representations. By constraining both representations to the same subspace, the knowledge from the source domain is effectively transferred to the target domain. Given the input data in the source and target domains $\mathbf{Y}_{S,i} \in \mathbb{R}^{m\times N_S}(i \in \{1,2\})$ and $\mathbf{Y}_{T,j} \in \mathbb{R}^{n\times N_T}(j \in \{1,2\})$, where m and n are the dimensions of the input data in the source and target domains and N_S and N_T denote the numbers of samples of the source and target domains, respectively, the SDCO model is defined as

$$\begin{cases} \mathbf{V}_{S,i} = \operatorname{argmin}\|\mathbf{Y}_{S,i} - \mathbf{D}_S\mathbf{V}_{S,i}\|_F^2 + \lambda_{S,i} f_S\left(\mathbf{V}_{S,i}\right) \\ \mathbf{V}_{T,j} = \operatorname{argmin}\|\mathbf{Y}_{T,j} - \mathbf{D}_T\mathbf{V}_{T,j}\|_F^2 + \lambda_{T,j} f_T\left(\mathbf{V}_{T,j}\right) \end{cases} \tag{1}$$

where $\mathbf{D}_S \in \mathbb{R}^{m\times K}$ and $\mathbf{D}_T \in \mathbb{R}^{n\times K}$ are the shared dictionaries of the source and target domains, respectively, $\mathbf{V}_{S,i} \in \mathbb{R}^{N_S\times K}$ and $\mathbf{V}_{T,j} \in \mathbb{R}^{N_T\times K}$ are the learned feature representations of $\mathbf{H}_{S,i}$ and $\mathbf{H}_{T,j}$ in the shared subspace, respectively, K denotes the size of the dictionaries, $\lambda_{S,i}$ and $\lambda_{T,j}$ are the regularization parameters in the source and target domains, and $f_S(\mathbf{V}_{S,i})$ and $f_T(\mathbf{V}_{T,j})$ are the regularization terms. The dictionary size is set smaller than the dimension of features ($N_S < K$, $N_T < K$), which ensures a low-dimension representation is learned to represent the original data.

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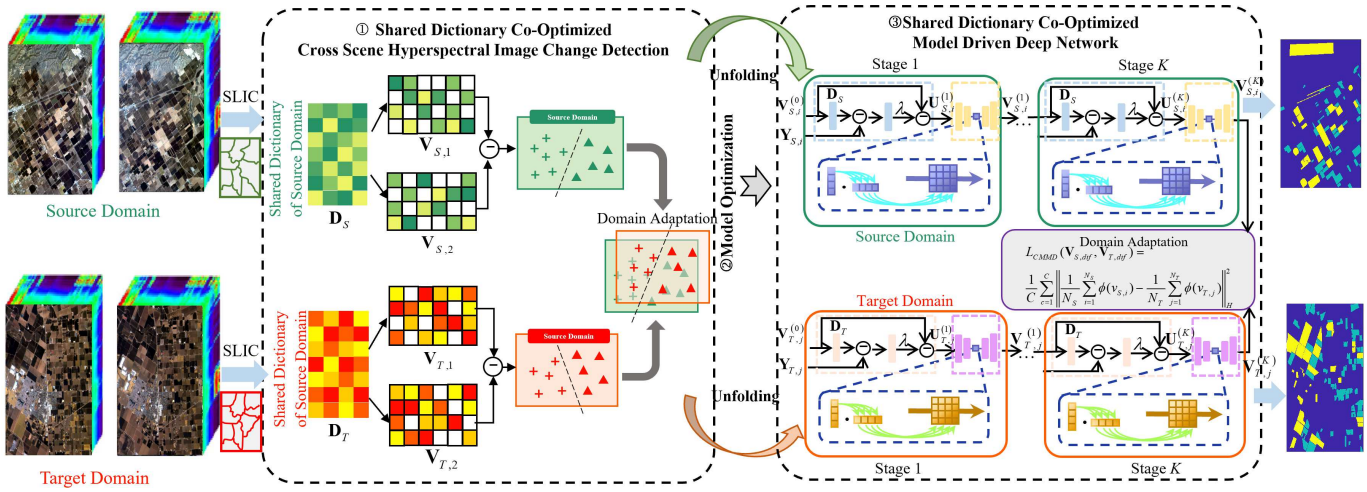


Fig. 1. Framework overview of the proposed SDCO-MDDN for cross scene HSI CD.

After constructing the SDCO model for cross scene HSI CD, we introduce an optimization algorithm to transform the model solving process of (1) into an iterative process of two subproblems. The deep neural network consisting of source and target branches with K stages is built with the guidance of the two subproblems, in which each module corresponds to the operation in the formula. In this way, the network takes the advantages of the model-based method to learn more effective representation of two domains in an interpretable manner. The representations of multitemporal HSIs of both domain $\mathbf{V}_{T,j}^{(K)}, j \in \{1,2\}$ and $\mathbf{V}_{S,i}^{(K)}, i \in \{1,2\}$ can be obtained after K iterations.

The difference features $\mathbf{V}_{T,\text{dif}}$ are acquired by performing subtraction operation on the learned representations of both branches in the target domain, which can be computed as

$$\mathbf{V}_{T,\text{dif}} = \mathbf{V}_{T,1}^{(K)} - \mathbf{V}_{T,2}^{(K)}. \tag{2}$$

Similarly, the difference feature $\mathbf{V}_{S,\text{dif}}$ of source domain is formulated as

$$\mathbf{V}_{S,\text{dif}} = \mathbf{V}_{S,1}^{(K)} - \mathbf{V}_{S,2}^{(K)}. \tag{3}$$

Then, the distribution of difference features of two domains is aligned to compel target domain to learn knowledge of the labeled samples in the source domain. Therefore, the proposed network trained with sufficient labeled samples in source domain and a few labeled samples in target domain can be used to detect changed areas of target domain.

To obtain the final CD results, the difference features are fed into fully connected layers followed by a softmax layer to identify the changed and unchanged superpixels. The final pixel-level detection result $\widehat{\mathbf{G}}_T$ can be transformed from superpixel-level result by superpixel query matrix $\mathbf{Q}_T$

$$\widehat{\mathbf{G}}_T = \mathbf{Q}_T^{\mathrm{T}} \cdot \text{softmax}\left(fc\left(\mathbf{V}_{T,\text{dif}}\right)\right) \tag{4}$$

where $fc(\cdot)$ is the fully connected layers. Similarly, we can obtain the final detection result of source domain $\widehat{\mathbf{G}}_S$ by

$$\widehat{\mathbf{G}}_S = \mathbf{Q}_S^{\mathrm{T}} \cdot \text{softmax}\left(fc\left(\mathbf{V}_{S,\text{dif}}\right)\right). \tag{5}$$

B. SDCO Model-Driven Deep Network

1) Model Optimization: The SDCO model is proposed to learn the representations of multitemporal HSIs of source and target domain in the same subspace jointly in Section III-A. To solve the proposed model by unfolding the inference, we adopt optimization algorithm to update the variables. Equation (1) encompasses both differentiable and nondifferentiable components. The inclusion of the nondifferentiable part necessitates the adoption of the PGD algorithm to effectively tackle this objective function. The expression for $\mathbf{V}_{T,j}^{(k+1)}(j \in \{1,2\})$ obtained at the $(t+1)$th iteration in the target domain is as follows:

$$\begin{aligned}\mathbf{V}_{T,j}^{(k+1)} &= \text{Prox}\left(\mathbf{V}_{T,j}^{(k)} - \lambda_{T,j}\nabla_{\mathbf{V}_T}\left(\frac{1}{2}\|\mathbf{Y}_{T,j} - \mathbf{D}_T\mathbf{V}_{T,j}^{(k)}\|_F^2\right)\right) \\ &= \text{Prox}\left(\mathbf{U}_{T,j}^{(k)}\right)\end{aligned} \tag{6}$$

where $\text{Prox}(\cdot)$ represents the proximal operator related to the non-smooth term and λ_T denotes the step size. The gradient of $1/2\|\mathbf{Y}_{T,j} - \mathbf{D}_T\mathbf{V}_{T,j}^{(k)}\|_F^2$ can be obtained as follows:

$$\begin{aligned}&\nabla_{\mathbf{V}_T}\left(\frac{1}{2}\|\mathbf{Y}_{T,j} - \mathbf{D}_T\mathbf{V}_{T,j}^{(k)}\|_F^2\right) \\ &= \nabla_{\mathbf{V}_T}\left(\frac{1}{2}\left(\text{tr}\left[\left(\mathbf{Y}_{T,j} - \mathbf{D}_T\mathbf{V}_{T,j}^{(k)}\right)^{\mathrm{T}}\left(\mathbf{Y}_{T,j} - \mathbf{D}_T\mathbf{V}_{T,j}^{(k)}\right)\right]\right)\right) \\ &= \mathbf{D}_T^{\mathrm{T}}\left(\mathbf{D}_T\mathbf{V}_{T,j}^{(k)} - \mathbf{Y}_{T,j}\right)\end{aligned} \tag{7}$$

where $\mathbf{D}_T$ represents the dictionary matrix of the target domain.

Therefore, in each iteration, $\mathbf{V}_{T,j}$ and $\mathbf{U}_{T,j}$ can be updated alternately

$$\mathbf{U}_{T,j}^{(k)} = \mathbf{V}_{T,j}^{(k)} - \lambda_{T,j}\left(\mathbf{D}_T^{\mathrm{T}}\left(\mathbf{D}_T\mathbf{V}_{T,j}^{(k)} - \mathbf{Y}_{T,j}\right)\right). \tag{8}$$

Subsequently, the proximal operator $\text{Prox}(\cdot)$ was applied to $\mathbf{U}_{T,j}^{(k)}$, denoted as

$$\mathbf{V}_{T,j}^{(k+1)} = \text{Prox}\left(\mathbf{U}_{T,j}^{(k)}\right). \tag{9}$$

Similarly, the objective function of the bi-temporal branch of the target domain can be obtained through the iterative (6), where $\mathbf{V}_{T,j}^{(k+1)}$ is the intermediate variable set

$$\begin{cases}\mathbf{U}_{T,j}^{(k)} = \mathbf{V}_{T,j}^{(k)} - \lambda_{T,j}\left(\mathbf{D}_T^{\mathrm{T}}\left(\mathbf{D}_T\mathbf{V}_{T,j}^{(k)} - \mathbf{Y}_{T,j}\right)\right) \\ \mathbf{V}_{T,j}^{(k+1)} = \text{Prox}\left(\mathbf{U}_{T,j}^{(k)}\right).\end{cases} \tag{10}$$

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Algorithm 1 Algorithm for the Proposed SDCO-MDDN

Require: HSIs and groud truth of the source domain $\mathbf{Y}_{S,1}$, $\mathbf{Y}_{S,2}$ and $\mathbf{G}_S$;
HSIs of the target domain: $\mathbf{Y}_{T,1}$, $\mathbf{Y}_{T,2}$;
number of epochs Max_epoch;
number of optimization iterations Max_iter;
initialize the SDCO-MDDN with parameters Θ;
Ensure: $\mathbf{G}_T$;
while $e \leq$ Max_epoch **do**
  $e \leftarrow 1$
  while $i \leq$ Max_iter **do**
    initialize $\mathbf{V}_s$ and $\mathbf{V}_t$;
    $i \leftarrow 1$
    update $\mathbf{U}_s$ and $\mathbf{U}_t$ according to Eq. (12) and (14);
    update $\mathbf{V}_s$ and $\mathbf{V}_t$ according to Eq. (20)-(25);
    $i \leftarrow i + 1$;
  end while
  obtain the result $\mathbf{G}_T$ and $\mathbf{G}_S$ according to the formulas (2)–(5);
  calculate total loss according to Eq. (26)-(33);
  update the model parameters Θ by optimizing $\mathcal{L}_{\text{total}}$;
  $e \leftarrow e + 1$;
end while
return $\mathbf{G}_T$;

The objective function in the source domain undergoes the same reasoning process as that in the target domain. The updating formulas for parameters $\mathbf{U}_{S,i}^{(k)}(i \in \{1,2\})$ and $\mathbf{V}_{S,i}^{(k+1)}(i \in \{1,2\})$ in the source domain are provided as

$$\begin{cases} \mathbf{U}_{S,i}^{(k)} = \mathbf{V}_{S,i}^{(k)} - \lambda_{S,i}\left(\mathbf{D}_S^{\mathrm{T}}\left(\mathbf{D}_S\mathbf{V}_{S,i}^{(k)} - \mathbf{Y}_{S,i}\right)\right) \\ \mathbf{V}_{S,i}^{(k+1)} = \text{Prox}\left(\mathbf{U}_{S,i}^{(k)}\right). \end{cases} \tag{11}$$

2) Network Architecture: Traditional model-based methods require multiple iterations to converge, resulting in high computational cost. In this article, we propose an SDCO-MDDN by constructing two subnetworks to map the iteration process of (10) and (11). As shown in Fig. 1, the proposed SDCO-MMDN adopts convolution layers instead of matrix computation to model all operators in the model, thereby facilitating the establishment of an end-to-end network. Since each module in the network has clear physical meanings, it is easy to track and correct the uncertainties of the network. Furthermore, the proposed network jointly learns the representations of two domains in the same subspace by aligning the distribution of the representations in an interpretable manner, so as to fully exploit that of source domain for CD of target domain.

The overall architecture of SDCO-MDDN is shown in Fig. 2. The input multitemporal HSI of source and target domains is transformed into a shared feature space to learn the representations of the HSI jointly. The learned representations that are obtained by iteratively updating $\mathbf{U}$ and $\mathbf{V}$ can be regarded as the features of the inputs. Two subnetworks are constructed to iteratively learn $\mathbf{U}$ and $\mathbf{V}$, respectively. The proposed network has K stages, equivalent to the K iterations in the iterative algorithm for unfolding (10) and (11). Specifically, for the target domain, (10) is decomposed into the following basic operations:

$$\mathbf{X}_{T,j}^{(k)} = \mathbf{D}_T^{\mathrm{T}}\left(\mathbf{Y}_{T,j} - \mathbf{D}_T\mathbf{V}_{T,j}^{(k)}\right), \quad j = 1,2 \tag{12a}$$

$$\mathbf{E}_{T,j}^{(k)} = \lambda_{T,j}\mathbf{X}_{T,j}^{(k)}, \quad j = 1,2 \tag{12b}$$

$$\mathbf{U}_{T,j}^{(k)} = \mathbf{V}_{T,j}^{(k)} - \mathbf{E}_{T,j}^{(k)}, \quad j = 1,2 \tag{12c}$$

$$\mathbf{V}_{T,j}^{(k+1)} = \text{Prox}\left(\mathbf{U}_{T,j}^{(k)}\right), \quad j = 1,2. \tag{13}$$

Similarly, for the source domain, (11) is decomposed into the following basic operations:

$$\mathbf{X}_{S,i}^{(k)} = \mathbf{D}_S^{\mathrm{T}}\left(\mathbf{Y}_{S,i} - \mathbf{D}_S\mathbf{V}_{S,i}^{(k)}\right), \quad i = 1,2 \tag{14a}$$

$$\mathbf{E}_{S,i}^{(k)} = \lambda_{S,i}\mathbf{X}_{S,i}^{(k)}, \quad i = 1,2 \tag{14b}$$

$$\mathbf{U}_{S,i}^{(k)} = \mathbf{V}_{S,i}^{(k)} - \mathbf{E}_{S,i}^{(k)}, \quad i = 1,2 \tag{14c}$$

$$\mathbf{V}_{S,i}^{(k+1)} = \text{Prox}\left(\mathbf{U}_{S,i}^{(k)}\right), \quad i = 1,2. \tag{15}$$

The detailed introduction of two subnetworks in both the two domains is as follows.

a) *Unfolding Subformulas (12) and (14):* The subnetwork updates the estimations $\mathbf{U}_{S,i}^{(t)}$ and $\mathbf{U}_{T,j}^{(t)}$ for the source and target domains, respectively. Since the multitemporal HSIs in each domain capture the same region, all pixels can be expressed as different linear combinations of the same components. Thus, a shared complete dictionary is learned to iteratively update $\mathbf{U}_{S,i}^{(t)}$ and $\mathbf{U}_{T,j}^{(t)}$.

In the target domain, the multitemporal HSIs are first segmented into superpixels and flattened into vectors. These vectors are passed through a convolutional layer to obtain the initial coefficient vector $\mathbf{V}_{T,j}^{(0)}$. As shown in Fig. 3, a 3×3 convolutional layer with stride 2 is then employed to learn the shared dictionary $\mathbf{D}_T$, with parameters shared to ensure consistency within the domain. The dictionary columns act as base atoms, extracting key features from the original HSI while reducing dimensionality. Finally, the output is subtracted from the input $\mathbf{Y}_{T,j}$ and convolved with the transpose of the learned dictionary to update $\mathbf{U}_{T,j}^{(t)}$, formulated as

$$\mathbf{X}_{T,j}^{(k)} = f_{\text{cov},\mathbf{D}_T^{\mathrm{T}}}\left(\mathbf{Y}_{T,j} - f_{\text{cov},\mathbf{D}_T}\left(\mathbf{V}_{T,j}^{(k)}\right)\right), \quad j = 1,2 \tag{16}$$

where $f_{\text{cov},\mathbf{D}_T}(\cdot)$ and $f_{\text{cov},\mathbf{D}_T^{\mathrm{T}}}$ refer to a convolutional layer that learns the dictionary matrix $\mathbf{D}_T$ and the transposed dictionary matrix, respectively, and $(\cdot)^{\mathrm{T}}$ refers to the transpose operation.

Based on (12b) and (12c), we devise a mechanism to adjust features $\mathbf{X}_{T,j}^{(k)}$ using a learnable parameter $\lambda_{T,j}$ and ultimately fuse $\mathbf{X}_{T,j}^{(k)}$ with coefficient matrix $\mathbf{V}_{T,j}^{(k)}$, which can be expressed as

$$\mathbf{U}_{T,j}^{(t)} = \mathbf{V}_{T,j}^{(k)} - \lambda_{T,j}\mathbf{X}_{T,j}^{(k)}, \quad i = 1,2 \tag{17}$$

where $\lambda_{T,j}$ is the penalty parameter.

For the source domain, following (14), the subnetwork aims to generate the intermediate variable $\mathbf{U}_{S,i}^{(k)}$ similarly as the target domain

$$\mathbf{X}_{S,i}^{(k)} = f_{\text{cov},\mathbf{D}_S^{\mathrm{T}}}\left(\mathbf{Y}_{S,i} - f_{\text{cov},\mathbf{D}_S}\left(\mathbf{V}_{S,i}^{(k)}\right)\right), \quad i = 1,2 \tag{18}$$

$$\mathbf{U}_{S,i}^{(k)} = \mathbf{V}_{S,i}^{(k)} - \lambda_{S,i}\mathbf{X}_{S,i}^{(k)}, \quad i = 1,2 \tag{19}$$

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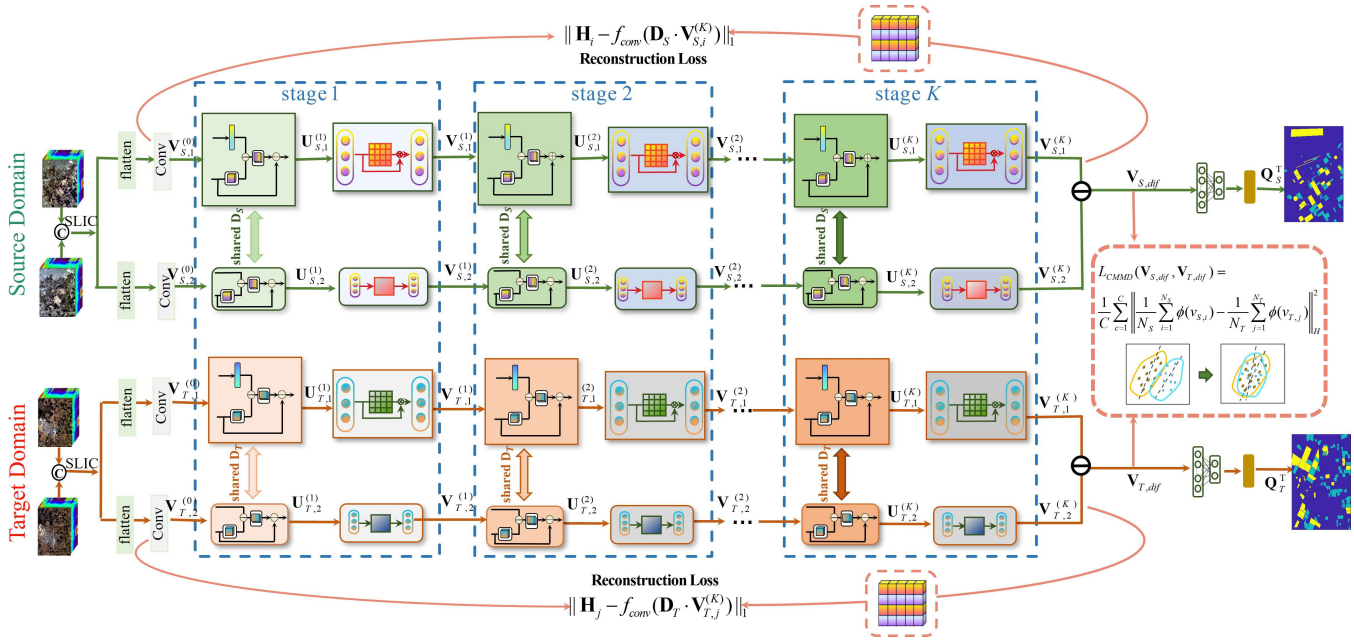


Fig. 2. Overall architecture of the proposed SDCO-MDDN.

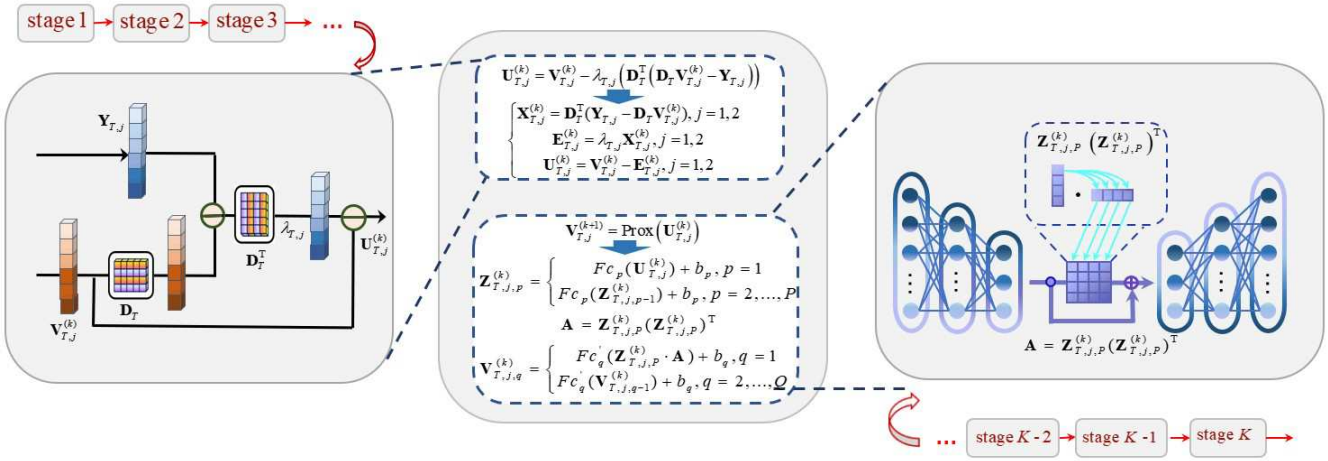


Fig. 3. Detailed architecture of the subnetworks in the proposed SDCO-MDDN.

where $f_{\mathrm{cov},\mathbf{D}_S}(\cdot)$ and $f_{\mathrm{cov},\mathbf{D}_S^{\mathrm{T}}}$ denote a convolutional layer that learns the dictionary matrix $\mathbf{D}_S$ and the transposed dictionary matrix of source domain, respectively.

C. Loss Function

b) *Unfolding Subformulas (13) and (15):* Utilizing proximal operators to incorporate prior information enables the imposition of prior constraints on optimization problems. Nevertheless, manually specified priors (e.g., sparsity [38] and low-rank [39]) often rely on relatively simple or direct formulations, thus failing to fully capture the complex structures and latent correlations present in the data. In order to learn the prior knowledge from the input data automatically, we construct a subnetwork with encoder–decoder architecture as backbone. However, there is a bottleneck for the encoder–decoder architecture to capture global information of the input feature. To solve this problem, we proposed a global self-correlation matrix that allows capturing relationships between each superpixel and others of the whole input data, thereby comprehensively considering the correlation of the entire graph. Therefore, by enhancing the encoded feature with the proposed global self-correlation matrix, the subnetwork is able to capture global correlation to focus on the information with high correlation.

The subnetwork structure for updating $\mathbf{V}_{T,i}^{(k+1)}$ in the target domain is shown in Fig. 3. The encoder part of the subnetwork consists of three convolutional layers with a kernel size of 3 and the pth layer of the encoder can be described as

$$\mathbf{Z}_{T,j,p}^{(k)} = \begin{cases} Fc_p\left(\mathbf{U}_{T,j}^{(k)}\right) + b_p, & p = 1 \\ Fc_p\left(\mathbf{Z}_{T,j,p-1}^{(k)}\right) + b_p, & p = 2, \ldots, P \end{cases} \tag{20}$$

where $\mathbf{Z}_{T,j,p}^{(k)}$ represents the output of the pth layer in the encoder, P is the number of the layer of encoder, $Fc_p(\cdot)$

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represents the pth fully connected layer with nonlinear ReLU, and b_p refers to the bias vector.

Due to the limited receptive field of the convolutional layers, it is difficult for the subnetwork to capture the global information. The global self-correlation matrix is designed to enhance the ability to extract the global information of the subnetwork. The global self-correlation matrix $\mathbf{A}_T$ is obtained by multiplying the output of encoder $\mathbf{Z}_{T,j,P}^{(k)}$ with the transpose matrix of its own, each row of which quantifies the degree of correlation between the superpixel and others. The proposed global self-correlation matrix $\mathbf{A}_T$ encourages the network to focus on highly correlated information and filter out irrelevant information, which can be expressed as

$$\mathbf{A}_T = \mathbf{Z}_{T,j,P}^{(k)} \left(\mathbf{Z}_{T,j,P}^{(k)}\right)^{\mathrm{T}}. \tag{21}$$

The decoder part, which takes the enhanced feature as input, consists of three 3×3 deconvolutional layers with stride 2. The output of the qth layer of the decoder can be described as

$$\mathbf{V}_{T,j,q}^{(k)} = \begin{cases} Fc_q'\left(\mathbf{Z}_{T,j,P}^{(k)} \cdot \mathbf{A}_T\right) + b_q, & q = 1 \\ Fc_q'\left(\mathbf{V}_{T,j,q-1}^{(k)}\right) + b_q, & q = 2, \ldots, Q \end{cases} \tag{22}$$

where $Fc_q'(\cdot)$ represents the qth fully connected layer with ReLU, and Q is the number of the layer of decoder. The output of the Qth layer of the decoder $\mathbf{V}_{T,i,Q}^{(k)}$ is the output of the subnetwork of the kth iteration $\mathbf{V}_{T,i}^{(k+1)}$.

Similar to the target domain, the subnetwork of the source domain is designed to model the proximal operation to update $\mathbf{V}_{S,i}^{(k+1)}$

$$\mathbf{Z}_{S,i,p}^{(k)} = \begin{cases} Fc_p\left(\mathbf{U}_{S,i}^{(k)}\right) + b_p, & p = 1 \\ Fc_p\left(\mathbf{Z}_{S,i,p-1}^{(k)}\right) + b_p, & p = 2, \ldots, P \end{cases} \tag{23}$$

$$\mathbf{A}_S = \mathbf{Z}_{S,i,P}^{(k)} \left(\mathbf{Z}_{S,i,P}^{(k)}\right)^{\mathrm{T}} \tag{24}$$

$$\mathbf{V}_{S,i,q}^{(k)} = \begin{cases} Fc_q'\left(\mathbf{Z}_{S,i,P}^{(k)} \cdot \mathbf{A}_S\right) + b_q, & q = 1 \\ Fc_q'\left(\mathbf{V}_{S,i,q-1}^{(k)}\right) + b_q, & q = 2, \ldots, Q \end{cases} \tag{25}$$

where $\mathbf{Z}_{S,i,p}^{(k)}$ represents the output of pth layer of the encoder, and $\mathbf{V}_{S,i,q}^{(k)}$ represents the output of qth layer of the decoder. The output of the Qth layer of the decoder $\mathbf{V}_{S,i,Q}^{(k)}$ is the output of the subnetwork of the kth iteration $\mathbf{V}_{S,i}^{(k+1)}$.

We design an interpretable cross-scene CD loss function to optimize the proposed network. It consists of four parts: classification loss, reconstruction loss, interdomain difference alignment loss, and intradomain class separation loss. As CD is essentially a binary classification task, the classification loss in the target domain is computed as the cross-entropy between the predicted change map and $\mathbf{G}_T$, formulated as

$$L_T^{\mathrm{cls}} = \frac{1}{N_T} \sum \mathbf{G}_T \log\left(\widehat{\mathbf{G}}_T\right). \tag{26}$$

Similarly, the classification loss of source domain can be expressed as

$$L_S^{\mathrm{cls}} = \frac{1}{S_T} \sum \mathbf{G}_S \log\left(\widehat{\mathbf{G}}_S\right) \tag{27}$$

where $\mathbf{G}_S$ is the ground-truth map of the source domain.

To learn an effective coefficient matrix, we employ the reconstruction loss LT^{rec}, which minimizes the difference between the reconstructed data and the input. The reconstruction is obtained by multiplying the coefficient matrix $\mathbf{V}^{(K)}T, j$ with the dictionary $\mathbf{D}_T$ followed by convolution operations. The reconstruction loss of the target domain is defined as

$$L_{T,j}^{\mathrm{rec}} = \left\|\mathbf{H}_j - f_{\mathrm{conv}}\left(\mathbf{D}_T \cdot \mathbf{V}_{T,j}^{(K)}\right)\right\|_1, \quad j = 1, 2. \tag{28}$$

Similarly, the reconstruction loss of source domain is formulated as

$$L_{S,i}^{\mathrm{rec}} = \left\|\mathbf{H}_i - f_{\mathrm{conv}}\left(\mathbf{D}_S \cdot \mathbf{V}_{S,i}^{(K)}\right)\right\|_1, \quad i = 1, 2. \tag{29}$$

To mitigate the effects of domain differences, it is essential to align the distributions of both classes of features (changed and unchanged features) between source and target domains. Therefore, we employ the conditional MMD (CMMD) to measure the difference in distribution between two domains, forcing the network to learn identical changed and unchanged features of both domains. The interdomain difference alignment loss measures the mean discrepancy between the difference feature of two domains in the reproducing kernel Hilbert space

$$L_{\mathrm{CMMD}}^{\mathrm{inter}}\left(\mathbf{V}_{S,\mathrm{dif}}, \mathbf{V}_{T,\mathrm{dif}}\right) = \frac{1}{C}\sum_{c=1}^{C}\left\|\frac{1}{N_S}\sum_{i=1}^{N_S}\phi\left(v_{S,i}\right) - \frac{1}{N_T}\sum_{j=1}^{N_T}\phi\left(v_{T,j}\right)\right\|_H^2 \tag{30}$$

where $\phi(\cdot)$ represents the layer parameters of both domains transformed into embedded spaces, and C equals 2, representing variable and constant scenarios.

In order to ensure the separability of the changed and unchanged features in each domain, the intradomain separation loss is designed to utilize MMD to maximize the distribution of changed and unchanged difference features. The intradomain separation loss of target domain is formulated as

$$L_{T,\mathrm{MMD}}^{\mathrm{intra}}\left(\mathbf{V}_{T,\mathrm{dif}}^{uc}, \mathbf{V}_{T,\mathrm{dif}}^{c}\right) = \exp\left(-\left\|\frac{1}{N_T^{uc}}\sum_{i=1}^{N_T^{uc}}\phi\left(v_{T,i}^{uc}\right) - \frac{1}{N_T^{c}}\sum_{j=1}^{N_T^{c}}\phi\left(v_{T,j}^{c}\right)\right\|_H^2\right) \tag{31}$$

where $\mathbf{V}_{T,\mathrm{dif}}^{uc}$ and $\mathbf{V}_{T,\mathrm{dif}}^{c}$ are the changed and unchanged difference features in target domain. Similarly, the intradomain separation loss of source domain is formulated as

$$L_{S,\mathrm{MMD}}^{\mathrm{intra}}\left(\mathbf{V}_{S,\mathrm{dif}}^{uc}, \mathbf{V}_{S,\mathrm{dif}}^{c}\right) = \exp\left(-\left\|\frac{1}{N_S^{uc}}\sum_{i=1}^{N_S^{uc}}\phi\left(v_{S,i}^{uc}\right) - \frac{1}{N_S^{c}}\sum_{j=1}^{N_S^{c}}\phi\left(v_{S,j}^{c}\right)\right\|_H^2\right). \tag{32}$$

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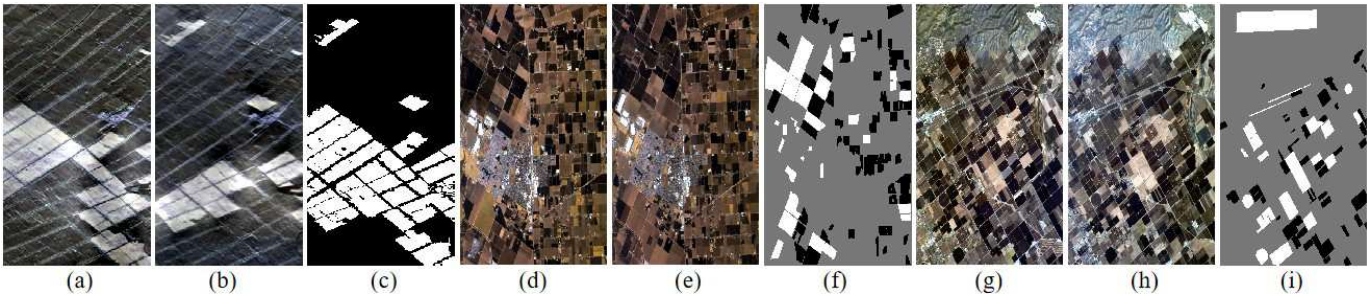


Fig. 4. HSI-CD datasets. (a)–(c) T1, T2, and GT of the China dataset. (d)–(f) T1, T2, and GT of the Bay Area dataset. (g)–(i) T1, T2, and GT of the Santa Barbara dataset.

TABLE I
DETAILED INFORMATION OF THE THREE DATASETS

Datasets	China	Bay Area	Santa Barbara
Size	420×140	600×500	984 × 740
Band	154	224	224
Date T_1	May 3, 2006	2013	2013
Date T_2	April 23, 2007	2015	2014
Site	Jiangsu Province, China	Patterson, USA	Santa Barbara, USA

The total interpretable cross scene CD loss of the network is formulated as

$$L_{\text{total}} = L_S^{\text{cls}} + L_T^{\text{cls}} + \alpha \left(\sum_{i=1}^{2} L_{S,i}^{\text{rec}} + \sum_{j=1}^{2} L_{T,j}^{\text{rec}} \right) + \beta L_{\text{CMMD}}^{\text{inter}} - \gamma \left(L_{S,\text{MMD}}^{\text{intra}} + L_{T,\text{MMD}}^{\text{intra}} \right) \tag{33}$$

where α, β, and γ are the tradeoff parameters (we set α, β, and γ as 0.01 in this article).

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we evaluate the performance of SDCO-MDDN against the state-of-the-art methods across three adaptation scenarios. The effectiveness of SDCO-MDDN for HSI CD is validated by analyzing the results both visually and quantitatively.

A. Data Description and Implementation Details

In the experiments, we employ three HSI CD datasets, i.e., China dataset, Bay Area dataset, and Santa Barbara dataset for cross scene CD. The China dataset is captured by the EO-1 Hyperion sensor, and the spectral and spatial resolutions are 30 m and 10 nm, respectively. The Bay Area and Santa Barbara datasets are captured by the AVIRIS sensor covering wavelengths from 0.4 to 2.5 μm, each of which contains two HSI with a spatial resolution of 20 m. The detailed information of the three datasets is shown in Table I. The subjective images of the three datasets, the reference images of the changed regions, and the reference images of the unchanged regions are presented in Fig. 4. The experiments are conducted on three HSI CD scenarios, including Santa Barbara→China, Bay Area→Santa Barbara, and Santa Barbara→Bay Area to evaluate the effectiveness of the proposed method in scenarios with images captured by both the same and different sensors.

We implement the proposed network on the Pytroch framework with a NVIDIA GeForce GTX 3090 GPU. Adam optimizer is adopted to train the network. The number of training epoch is 1000 and learning rate is set to 1e-4.

B. Evaluation Measures and Competing Methods

To evaluate the effectiveness of the proposed SDCO-MDDN, we employ two commonly used metrics, namely, overall accuracy (OA) and kappa coefficient (kappa). OA is a measure of the percentage of pixels that are correctly classified. AA is defined as the average of the accuracies for the change and no-change classes. Kappa measures the agreement between the change map and the ground truth (GT) map and $F1$-score is the harmonic mean of precision and recall for the change class. In the experiments, we compared SDCO-MDDN with both traditional and deep learning methods, including PCA, RECNN [40], SST-Former [41], CBANet [34], HAFF [42], FGDANet [18], and DA-former [43]. To ensure experimental fairness, 100 labeled samples in the target domain are selected as training samples during the training phase of all the deep learning-based methods, and the parameters of the competing methods are adjusted for best performance. Since FGDANet and the proposed method adopts SLIC to segment HSIs into superpixels as an input and the pixels in the same superpixel are regarded to have the same label, so the proposed method is able to obtain the label of all pixels in the superpixel, which contain training data, and the labeled superpixels are fed to the proposed model for training.

C. Performance on Different Scenarios

1) Santa Barbara → China: Fig. 5 shows the change maps generated by different competing methods on the Santa Barbara dataset (source domain) and China dataset (target domain). The visual results highlight the advantages of our method over others. The discrepancies identified in the area delineated by the red box suggest that our proposed method demonstrates a lower missed (red) and false (green) detection rate compared to other methods, especially PCA, RECNN, and SST-Former. To further confirm the effectiveness of our proposed method, we present the quantitative results in Table II. PCA exhibits unsatisfactory performance in terms of OA, AA, kappa, and $F1$. With a kappa of 0.7854, the RECNN method has a high false alarm rate and fails to obtain satisfactory results. In the competing methods, deep learning methods

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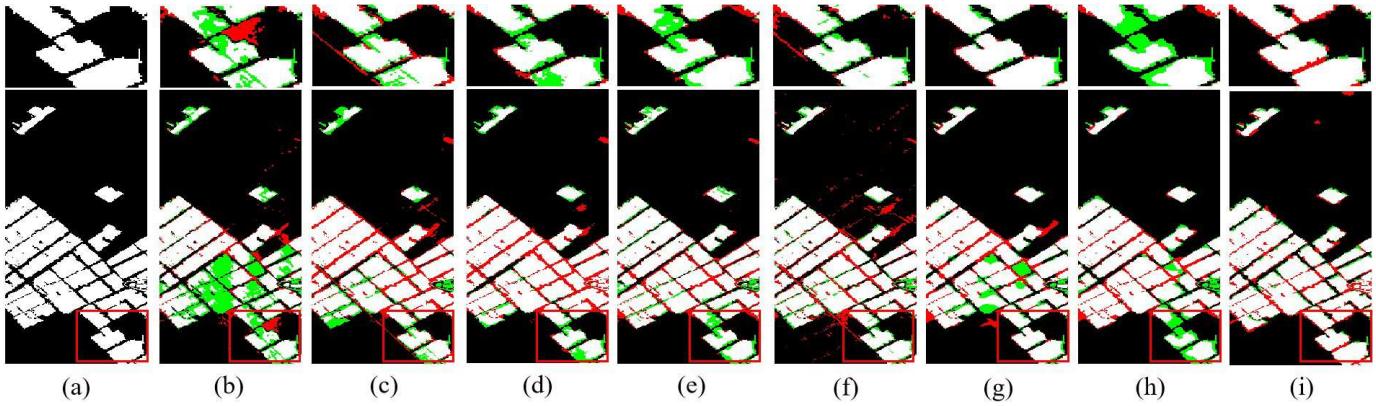


Fig. 5. Subjective results of different methods for Santa Barbara → China. (a) GT. (b) PCA. (c) RECNN. (d) SST-Former. (e) CBANet. (f) HAFF. (g) FGDANet. (h) DA-Former. (i) Proposed.

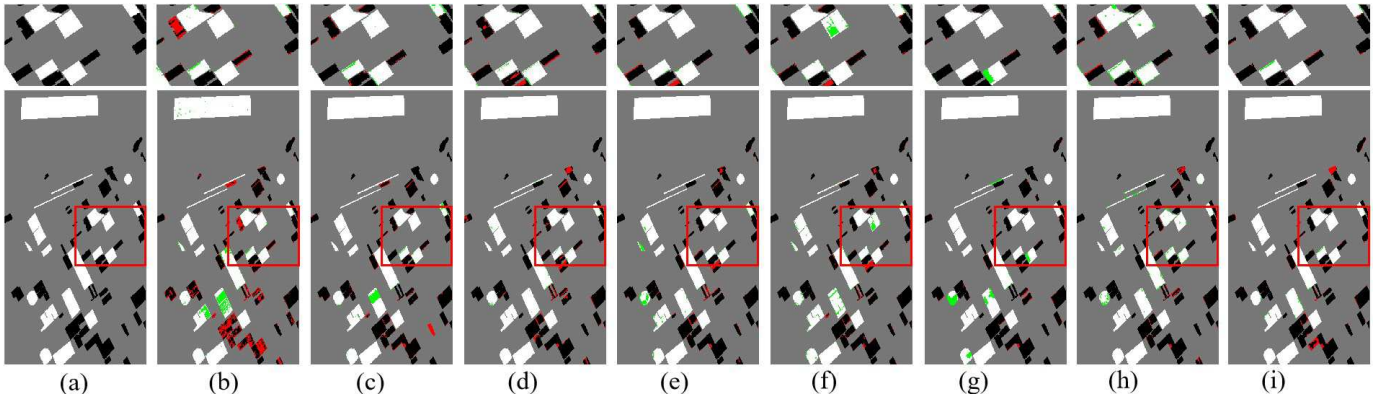


Fig. 6. Subjective results of different methods for Bay Area → Santa Barbara. (a) GT. (b) PCA. (c) RECNN. (d) SST-Former. (e) CBANet. (f) HAFF. (g) FGDANet. (h) DA-Former. (i) Proposed.

TABLE II
OBJECTIVE PERFORMANCE OF SDCO-MDDN AND OTHER COMPETING METHODS FOR SANTA BARBARA → CHINA DATASET

Methods	OA	AA	kappa	F1
PCA	0.8832	0.8403	0.7147	0.7954
RECNN	0.9012	0.9015	0.7854	0.8530
SST-Former	0.9233	0.9243	0.8262	0.8831
CBANet	0.9251	0.9179	0.8274	0.8823
HAFF	0.9257	0.9271	0.8317	0.8868
FGDANet	0.9219	0.9154	0.8205	0.8779
DA-former	0.9261	0.9082	0.8261	0.8793
Proposed	**0.9355**	**0.9344**	**0.8493**	**0.9015**

TABLE III
OBJECTIVE PERFORMANCE OF SDCO-MDDN AND OTHER COMPETING METHODS FOR BAY AREA → SANTA BARBARA DATASET

Methods	OA	AA	kappa	F1
PCA	0.8522	0.8310	0.6812	0.7956
RECNN	0.9483	0.9436	0.8911	0.9334
SST-Former	0.9583	0.9493	0.9118	0.9446
CBANet	0.9585	0.9544	0.9126	0.9466
HAFF	0.9556	0.9522	0.9068	0.9432
FGDANet	0.9577	0.9587	0.9119	0.9471
DA-former	0.9600	0.9580	0.9162	0.9491
Proposed	**0.9661**	**0.9588**	**0.9281**	**0.9555**

HAFF and DA-former show higher values in OA, AA, kappa, and $F1$. Compared to HAFF, our proposed approach shows improvements of 0.98%, 0.73%, 0.0176, and 0.0147 in OA, kappa, and $F1$, respectively. This is because the proposed method transfers the knowledge of the source domain in an interpretable way, which can learn the difference features in the target domain more effectively.

2) Bay Area → Santa Barbara: Fig. 6 shows the change maps of different methods on the Bay Area dataset (source domain) and Santa Barbara dataset (target domain). In subtle and dispersed areas, the traditional methods PCA demonstrates limited efficacy in detecting changes. In the CD maps produced by SST-Former, CBANet, and HAFF, numerous changed pixels are erroneously identified as unchanged pixels. The magnified regions of the CD maps for a and b exhibit more false detections. To comprehensively validate the efficacy of SDCO-MDDN, the objective experimental results are detailed in Table III. Table III shows that the OA, kappa, and $F1$ of PCA are particularly unsatisfactory. Although the accuracy of RECNN has improved compared to traditional methods, the value of kappa still does not exceed 0.9. SST-Former, CBANet, HAFF, FGDANet, and DA-former achieve

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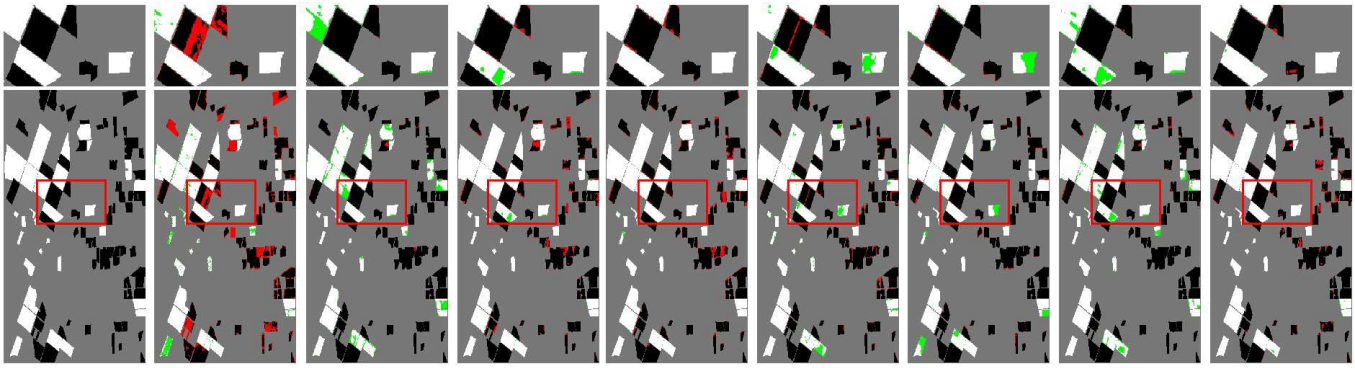


Fig. 7. Subjective results of different methods for Santa Barbara → Bay Area. (a) GT. (b) PCA. (c) RECNN. (d) SST-Former. (e) CBANet. (f) HAFF. (g) FGDANet. (h) DA-Former. (i) Proposed.

TABLE IV
OBJECTIVE PERFORMANCE OF SDCO-MDDN AND OTHER COMPETING METHODS FOR SANTA BARBARA → BAY AREA DATASET

Methods	OA	AA	kappa	F1
PCA	0.8564	0.8630	0.71.56	0.8509
RECNN	0.9447	0.9412	0.8881	0.9504
SST-Former	0.9528	0.9551	0.9056	0.9543
CBANet	0.9554	0.9580	0.9108	0.9566
HAFF	0.9540	0.9537	0.9075	0.9570
FGDANet	0.9590	0.9581	0.9175	0.9620
DA-former	0.9575	0.9554	0.9143	0.9612
Proposed	**0.9635**	**0.9655**	**0.9270**	**0.9648**

TABLE V
MODEL EFFICIENCY OF DIFFERENT CD METHODS FOR SANTA BARBARA → CHINA DATASET

Methods	FLOPs(G)	Para(M)	Time(s)
PCA	-	-	**0.9**
RECNN	1668.88	3.65	8.5
SST-Former	8539.00	2.50	10.7
CBANet	1096.05	0.33	8.2
HAFF	2964.58	2.67	6.2
FGDANet	0.8	0.18	13.4
DA-former	1785.42	0.40	9.7
Proposed	**0.44**	**0.038**	11.7

satisfactory performance, and the accuracy of DA-former is the highest with OA, AA, kappa, and $F1$ of 0.96, 0.9580, 0.9162, and 0.9491, respectively. The performance of the proposed SDCO-MDDN exceeds that of DA-former by 0.9661 in OA, 0.9588 in AA, 0.9281 in kappa, and 0.9555 in $F1$, underscoring the significance of domain adaptation in CD.

3) Santa Barbara → Bay Area: Fig. 7 presents the change map of the different methods and the proposed method on the Santa Barbara dataset (source domain) and Bay Area dataset (target domain). The visual result reveals that the change map generated by PCA is disappointing, containing numerous inaccurately detected areas. Among the deep learning methods, it is evident that SDCO-MDDN is the most consistent with the GT map, as further substantiated by performance metrics. The objective results of different methods are presented in Table IV. The values of OA, kappa, and $F1$ of PCA are lower than 0.9 as the worst performance. Although the performance of RECNN surpasses that of PCA, both OA and kappa of these methods are still suboptimal, indicating a significant disparity compared to our proposed method. The results of SST-Former, CBANet, and HAFF are better than other competing method, and the values of OA, AA, kappa, and $F1$ surpass 0.95, 0.95, 0.9, and 0.95, respectively. DAformer and FGDANet are cross-domain CD algorithms. They leverage the knowledge from the source domain, which effectively enhances the detection accuracy in the target domain. Our method attains the highest scores in OA, AA, kappa, and $F1$, further affirming its superior effectiveness in HSI CD.

D. Method Efficiency

Table V shows the FLOPs, parameters, and running times of different methods for Santa Barbara→China dataset. The deep learning-based methods have large FLOPs and are more time-consuming, but their performance is better than traditional methods. Overall, our algorithm manages to enhance the accuracy at an acceptable time cost and the small number of parameters. Since FGDANet and our method use superpixel segmentation, the computational load is the product of the number of superpixel points and the single run network cost. In contrast, other methods calculate based on the number of pixel points, which is usually much larger. Thus, FGDANet and our method have a lower computational complexity. The slightly longer running time is mainly due to the inherent cost of the superpixel segmentation step, which takes at least 9 s. However, this step significantly reduces the subsequent network computation and ensures superior detection accuracy, highlighting the efficiency–accuracy tradeoff advantage of our method. In future work, we will investigate more efficient processing strategies to further improve runtime.

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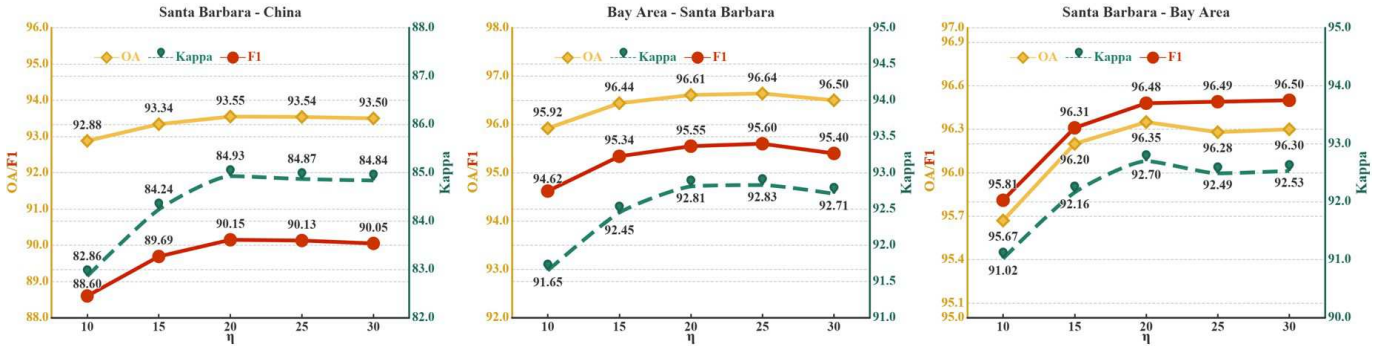


Fig. 8. Performance comparison of SDCO-MDDN with different numbers of dictionaries on three different scenarios.

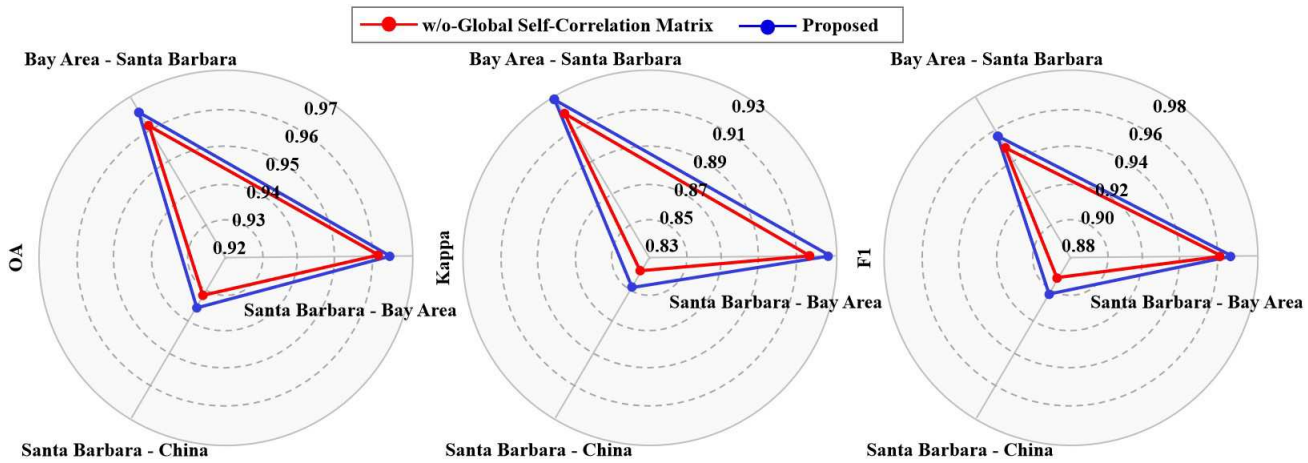


Fig. 9. Performance comparison of SDCO-MDDN and its variant, which removes global self-correlation matrix **A** (without-global self-correlation matrix) on three different scenarios.

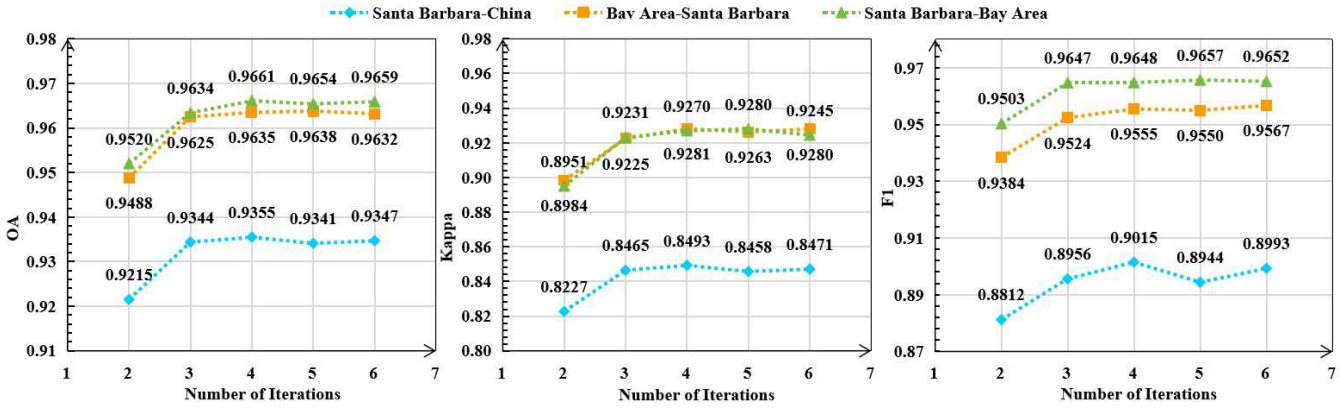


Fig. 10. Performance comparison of SDCO-MDDN with different numbers of iterations on three different scenarios.

E. Ablation Study

1) Number of Dictionary: In the proposed SDCO-MDDN, the HSI is decomposed into dictionaries and coefficients to reduce the dimension and learn the feature representation in the shared subspace. To investigate the effect of the number of dictionary, we carry out experiments to evaluate the capabilities of models with dictionary numbers to be 10, 15, 20, 25, and 30 on all the three scenarios. As shown in Fig. 8, the performance of SDCO-MDDN drops when dictionary size is limited, suggesting an oversimplified HSI representation. It can be observed that after the number of dictionary reaches 20, the performance starts to stabilize. Therefore, the number of dictionary is set to 20 in this article.

2) Effectiveness of Global Self-Correlation Matrix: In order to take the global spatial information of the HSI into consideration, the global self-correlation matrix is designed to aggregate relevant feature information. Fig. 9 shows the performance of SDCO-MDDN and its variant that removes the global self-correlation matrix (without-global self-correlation matrix) to evaluate the effectiveness of global self-correlation matrix. It is apparent that the utilization of SDCO-MDDN can make an increment in OA, kappa, and $F1$ on all the three scenarios. The results demonstrate that the global self-correlation matrix can fully exploit the global spatial information, which helps obtain better coefficient matrix for CD.

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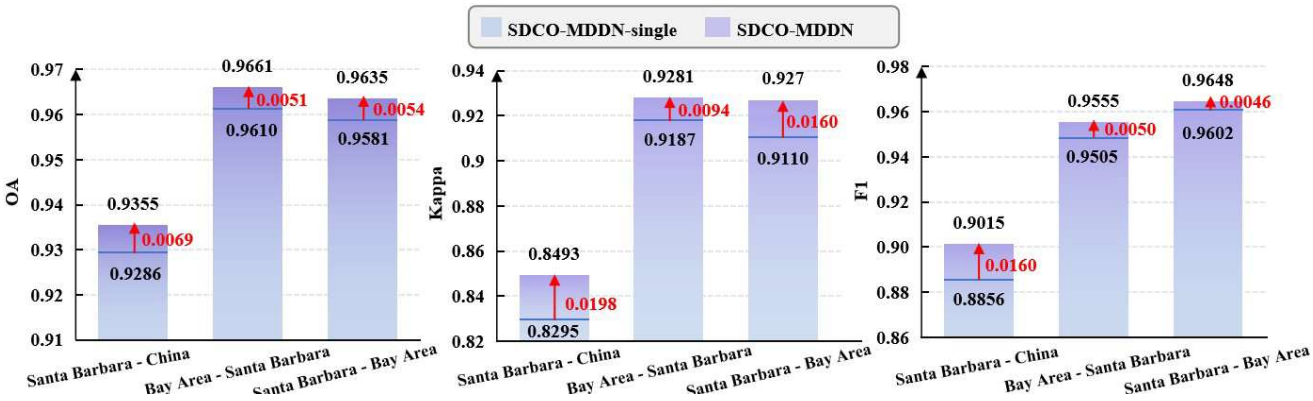


Fig. 11. Performance comparison of the SDCO-MDDN and its variant without domain adaptation (SDCO-MDDN-single) on three different scenarios.

3) Number of Iterations of the Network: An ablation study was conducted to determine the optimal SDCO-MDDN iterations on all the three scenarios. Fig. 10 shows the OA, kappa, and $F1$ of the results obtained by SDCO-MDDN with the number of iterations ranging from 2 to 6. It can be observed in Fig. 10 that the values of OA, kappa, and $F1$ of the model demonstrate a consistent improvement as the number of iterations increases from 1 to 4. However, the growth trend seems to level off after the number of iterations is larger than 4, and the additional number of iterations increases the computational cost. Based on the experimental results, we set the number of iterations as 4 to balance the performance and computational costs.

4) Effectiveness of Cross Scene Strategy: To demonstrate the effectiveness of cross scene strategy, we compare the results of SDCO-MDDN and its variant that only trains with labeled data in target domain without utilizing the knowledge of source domain (SDCO-MDDN-single). As shown in Fig. 11, the proposed network achieves significantly higher OA and kappa on three different scenarios compared to the SDCO-MDDN-single. Compared to SDCO-MDDN-single, the proposed method exhibits an increase in OA of 0.69%, kappa of 0.0198, and $F1$ of 0.016 on Santa Barbara→China datasets. This fully confirms that sufficient labeled data from the source domain can improve the detection accuracy in the target domain.

V. Conclusion

This article proposes an SDCO-MDDN for cross scene HSI CD. The core advancement of this study lies in building an interpretable cross scene network with the guidance of the proposed SDCO model to jointly learn the representations of HSI of source and target domains in an interpretable manner. Moreover, the global self-correlation matrix is designed to capture the global correlation information of the whole data, which help the network highlight the information with high correlation. The distribution of representations in different domains is aligned to minimize the domain shift, thus transferring knowledge of source domain for CD in target domain with limited labeled samples. The experimental results indicate that SDCO-MDDN excels beyond the most advanced CD methods in three distinct scenarios. Due to potential sensor attitude fluctuations and terrain irregularities, two-phase images may not register precisely. Inaccurate registration can distort target geometries, disrupt spatial correspondence, and likely cause many false detections in CD, reducing the procedure's reliability. So, in future work, we will focus on making our method more robust to misregistration for better real-world performance.

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